

IT3071 – Machine Learning and Optimization Methods

Introduction to Deep Learning

H.M. Samadhi Chathuranga Rathnayake

Classical Machine Learning VS Deep Learning

- Classical Machine Learning relies on manual feature extraction and simpler algorithms to make predictions.
- Deep Learning automatically learns features from raw data using multiple layers of neural networks, making it more powerful for complex tasks like image and speech recognition.
- Classical Machine Learning works like traditional programming with human-crafted features, while Deep Learning mimics the human brain by using artificial neural networks that automatically learn patterns.
- This biological inspiration allows Deep Learning to handle complex data like images and sounds more effectively.

Classical Machine Learning VS Deep Learning

Aspect	Classical Machine Learning	Deep Learning
Feature Engineering	Requires manual feature selection and engineering	Learns features automatically from raw data
Data Requirement	Performs well with smaller datasets	Requires large amounts of labeled data
Model Complexity	Models are generally simpler (Ex- decision trees, SVM, logistic regression)	Models are complex with many layers (e.g., deep neural networks)
Interpretability	Easier to interpret and explain	Harder to interpret (often considered “black-box” models)

Classical Machine Learning VS Deep Learning

Aspect	Classical Machine Learning	Deep Learning
Computational Power	Works on standard CPUs	Requires GPUs/TPUs for efficient training
Training Time	Generally faster to train	Takes longer due to deeper architectures
Application Areas	Tabular data, structured data tasks	Image, audio, video, NLP, and other unstructured data tasks
Dependency on Domain Knowledge	High – domain expertise often needed for feature design	Low – learns features automatically from data

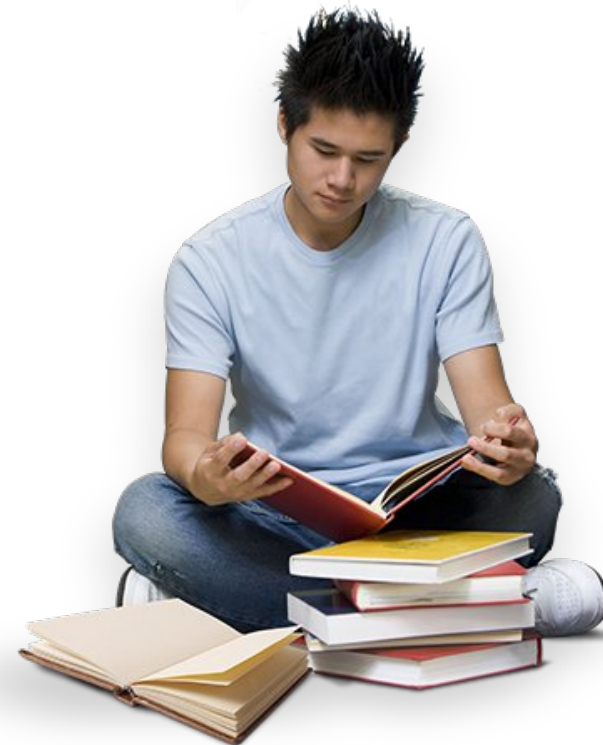
Deep Learning

- Brain inspired systems which are intended to replicate the way that we humans learn.
- Consist of input and output layers, as well as (in most cases) a hidden layer consisting of units that transform the input into something that the output layer can use.
- When a child is born, what does the child know? To our knowledge, the child knows only how to cry The child probably does not know its parents.
- When the child grows, the step by step learning process begins First, the child learns to drink milk
- Then the child learns to identify its parents Every time a child learns something, it is encoded into some portion of the brain.
- If we do not practice what we learned, we start to forget Consequently, by practice or training, we can hard code some selected things into our brains.



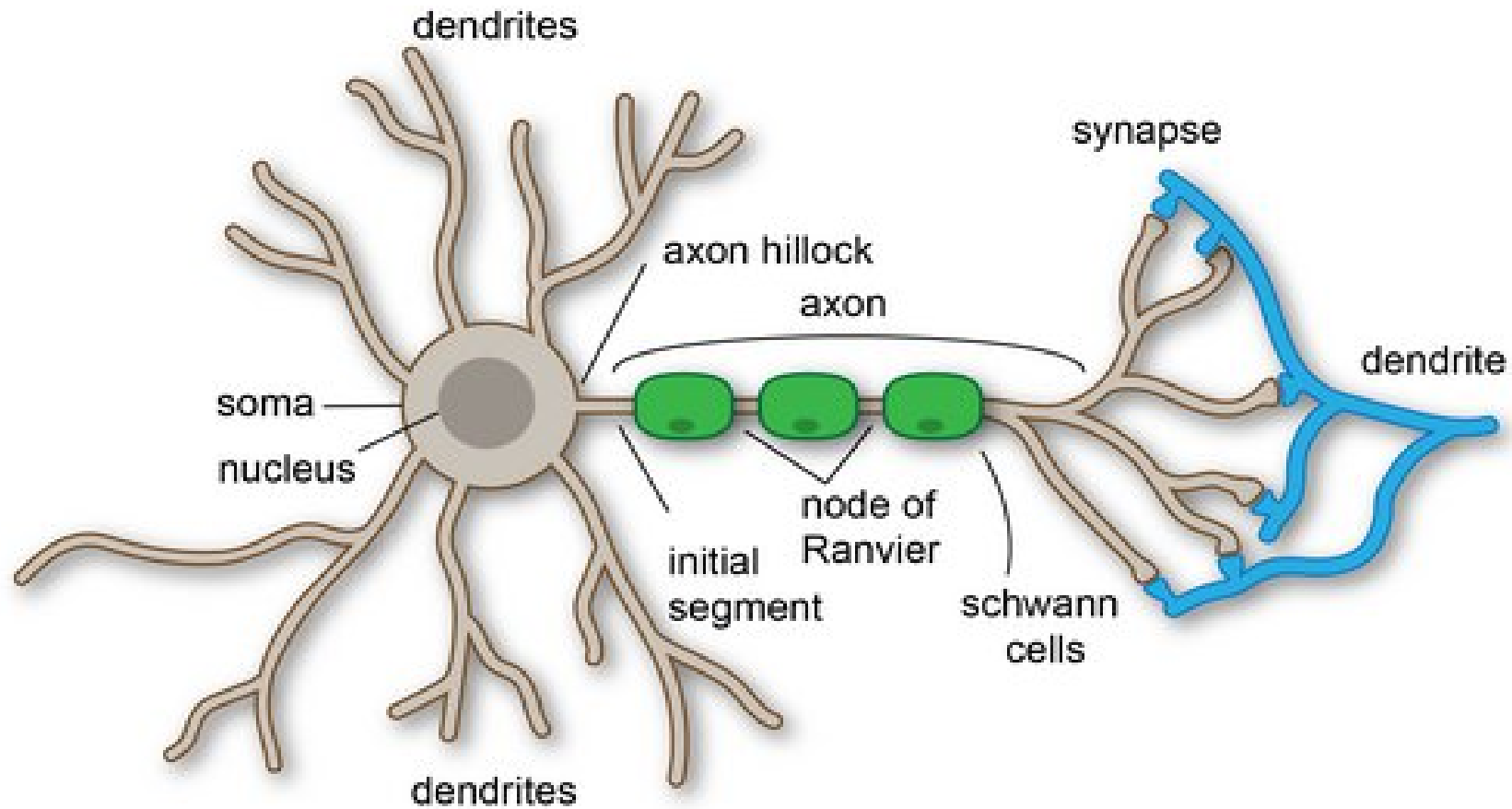
Deep Learning

- Neuroscientists believe that learning stimulates new dendrite connections between neurons.
- Greater usage of the brain through learning and stimulation creates greater dendrite connectivity thus, as we learn more and more, we become more intelligent.
- Wisdom is not created through genetics Wisdom and knowledge are based on how we learn and how we practice what we learned.



Perceptron

This is the general process of decision making inside a biological neuron.



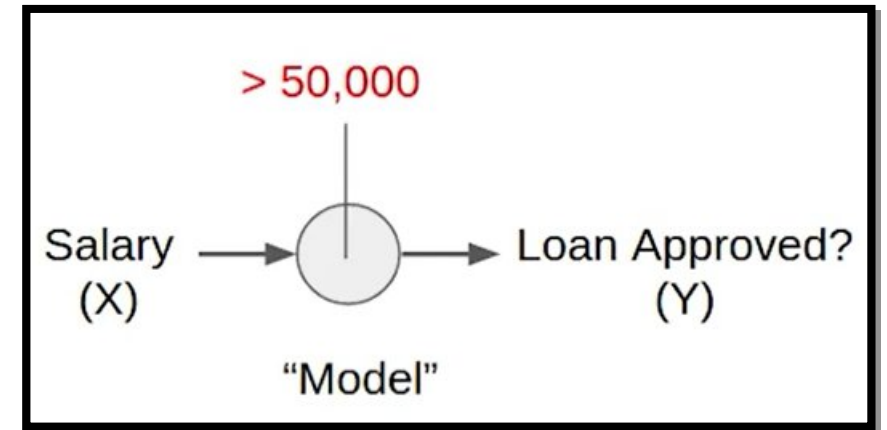
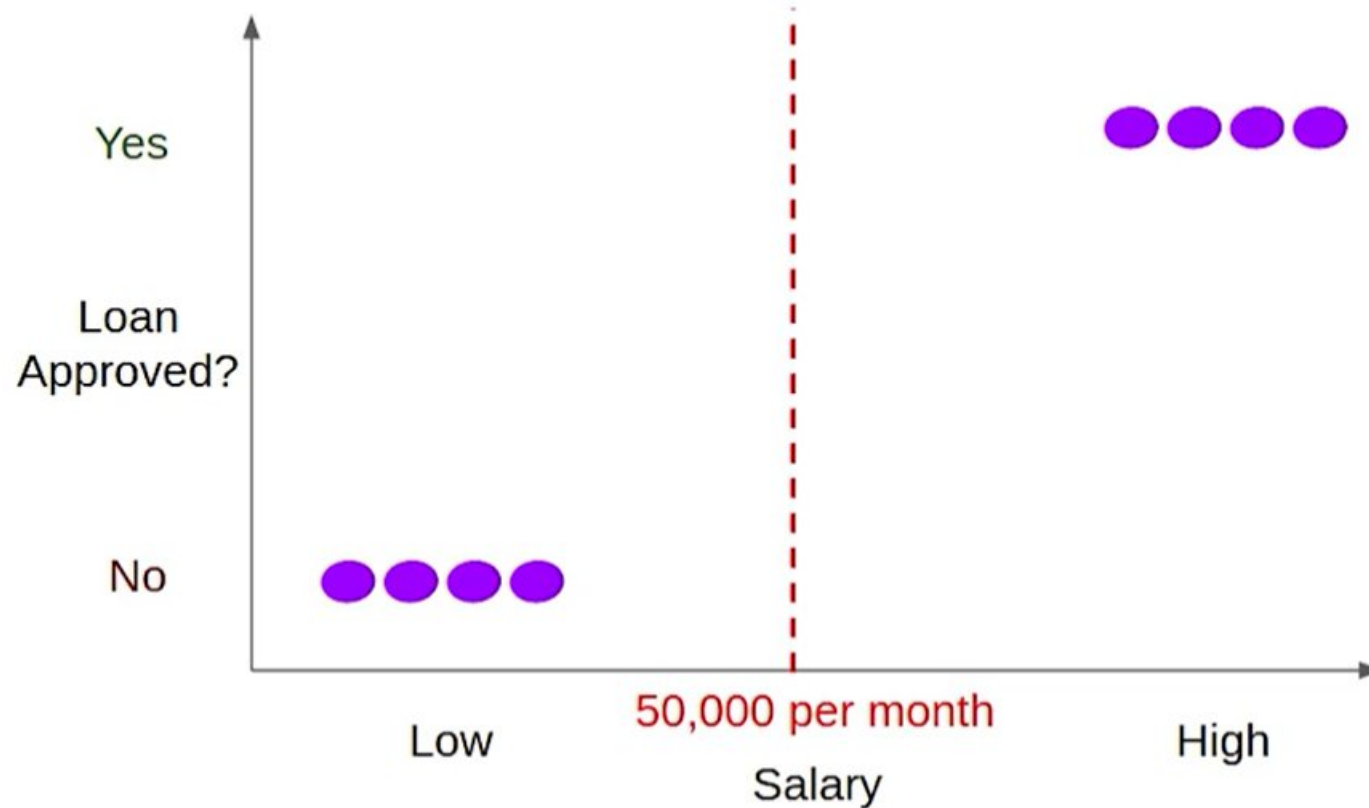
Perceptron

Think about following question. Here with the salary, it will check whether the loan is approved or not.



Perceptron

Think that the loan is going to approved when the salary is greater than 50000.

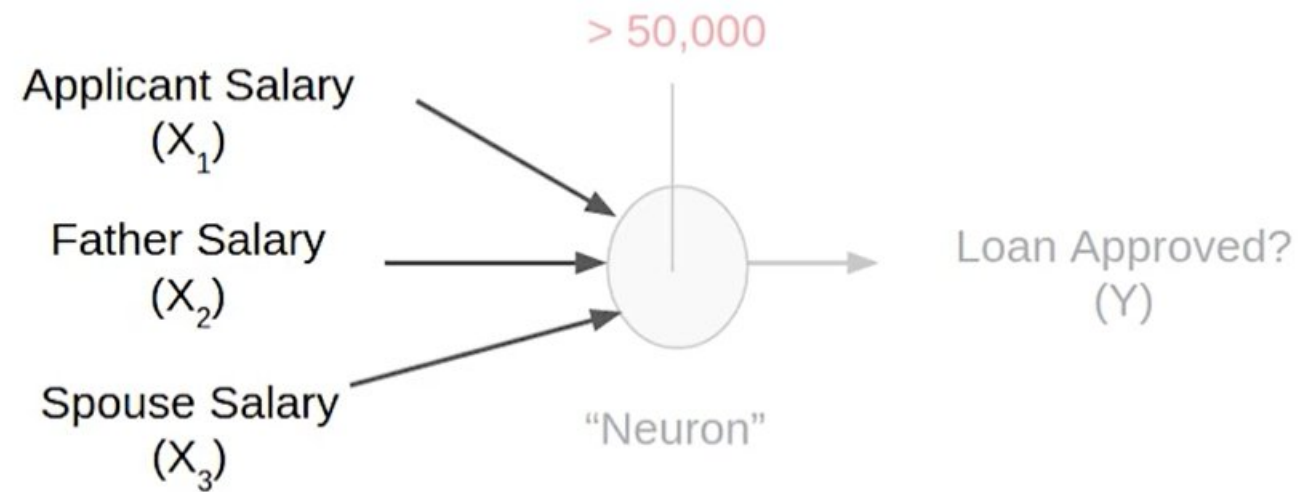


Tasks:

- Salary as input
- Check if it is at least 50000 or not?
- If the condition is True the approve the loan

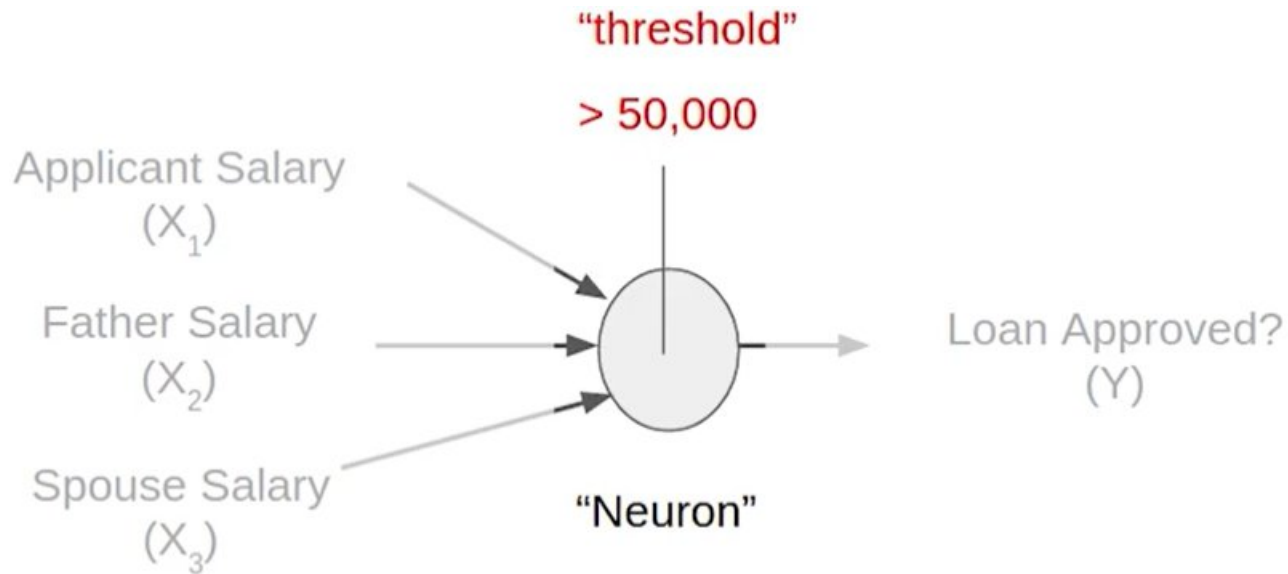
Perceptron

More than one input variables can be considered.



Perceptron

Consider that the addition of these input values is going to be compared with the threshold.



$$X_1 + X_2 + X_3 > T$$

$$X_1 + X_2 + X_3 - T > 0$$

$$X_1 + X_2 + X_3 + Bias > 0$$

$$\text{Total Income} = \text{Applicant Salary } (X_1) + \text{Father Salary } (X_1) + \text{Spouse Salary } (X_3)$$

Total Income > threshold ?

$$X_1 + X_2 + X_3 + Bias > 0$$

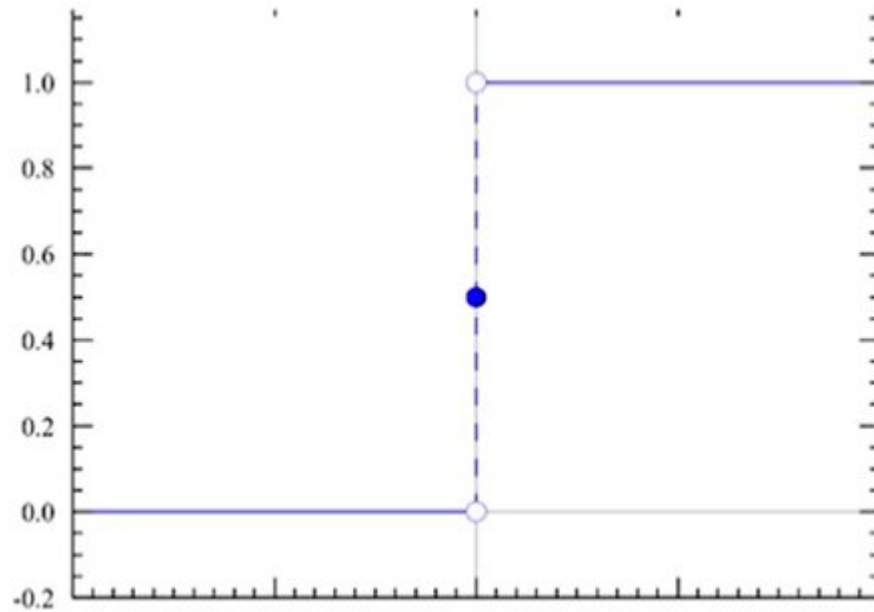
1

$$X_1 + X_2 + X_3 + Bias < 0$$

0

Perceptron

This follows the step function.

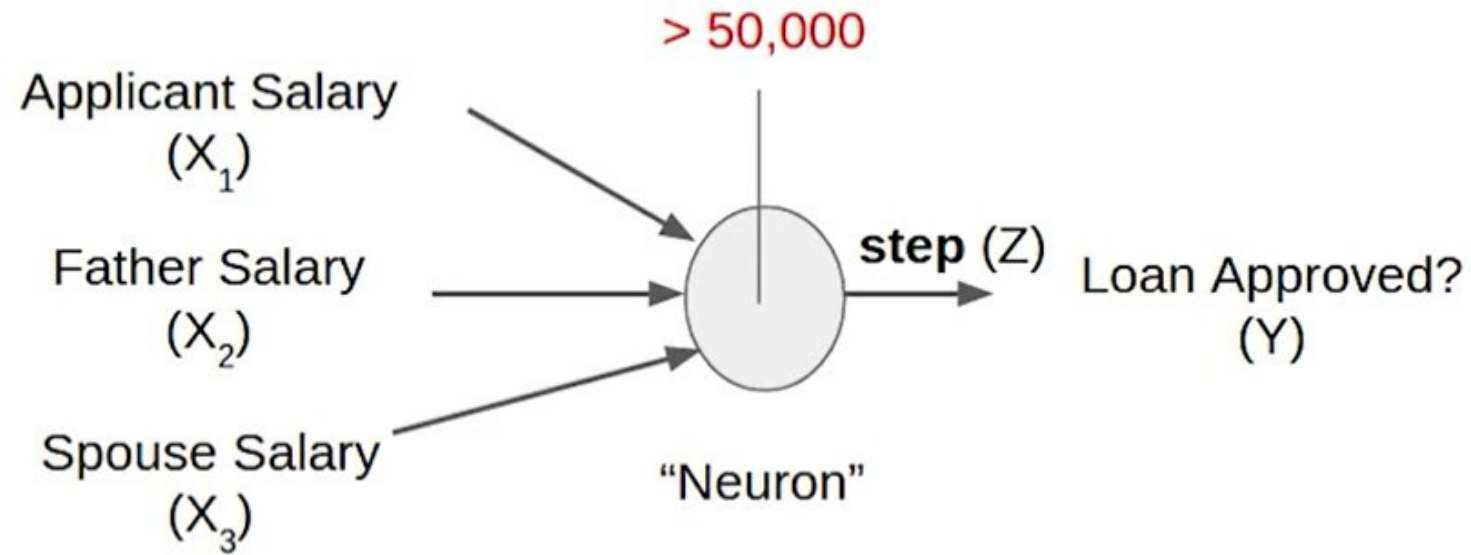


$$Z = X_1 + X_2 + X_3 + Bias$$

$$Step(Z) = Y = \begin{cases} 1 & Z > 0 \\ 0 & Z \leq 0 \end{cases}$$

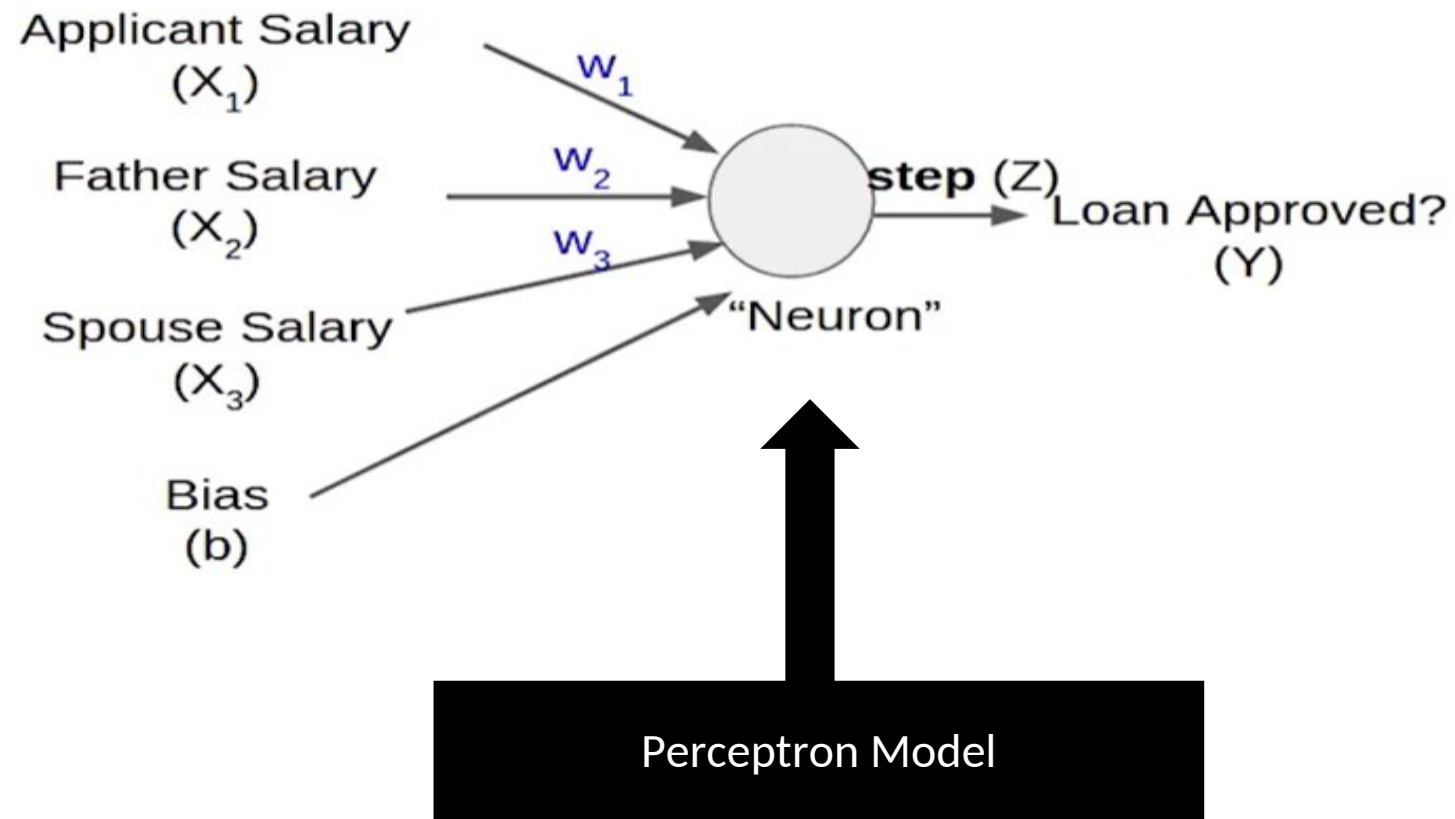
Perceptron

Finally, the system can be obtained as,



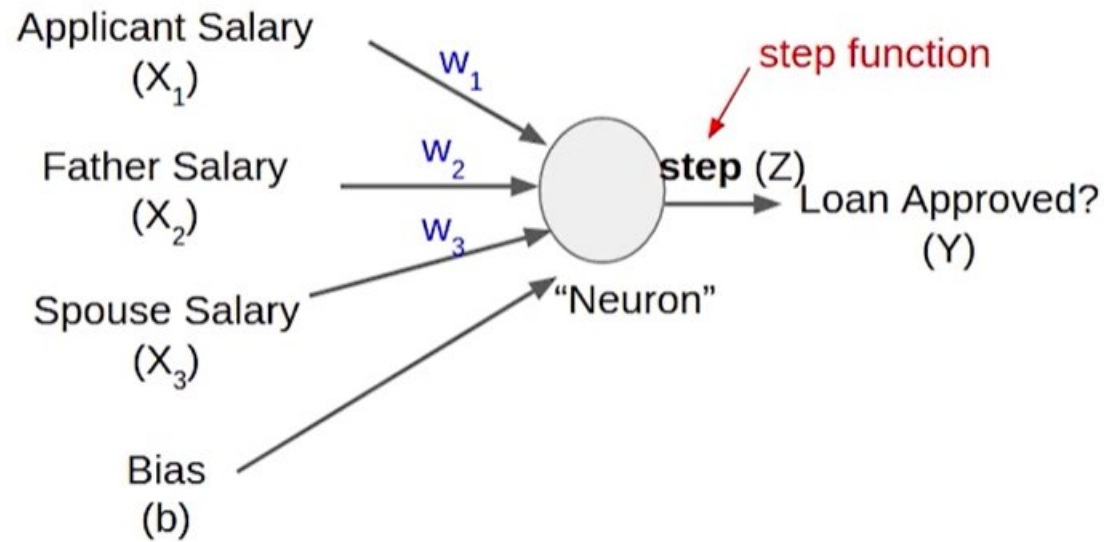
Perceptron

The weight can be changed for each input.



Perceptron

Weights and bias values can be changed and updated.



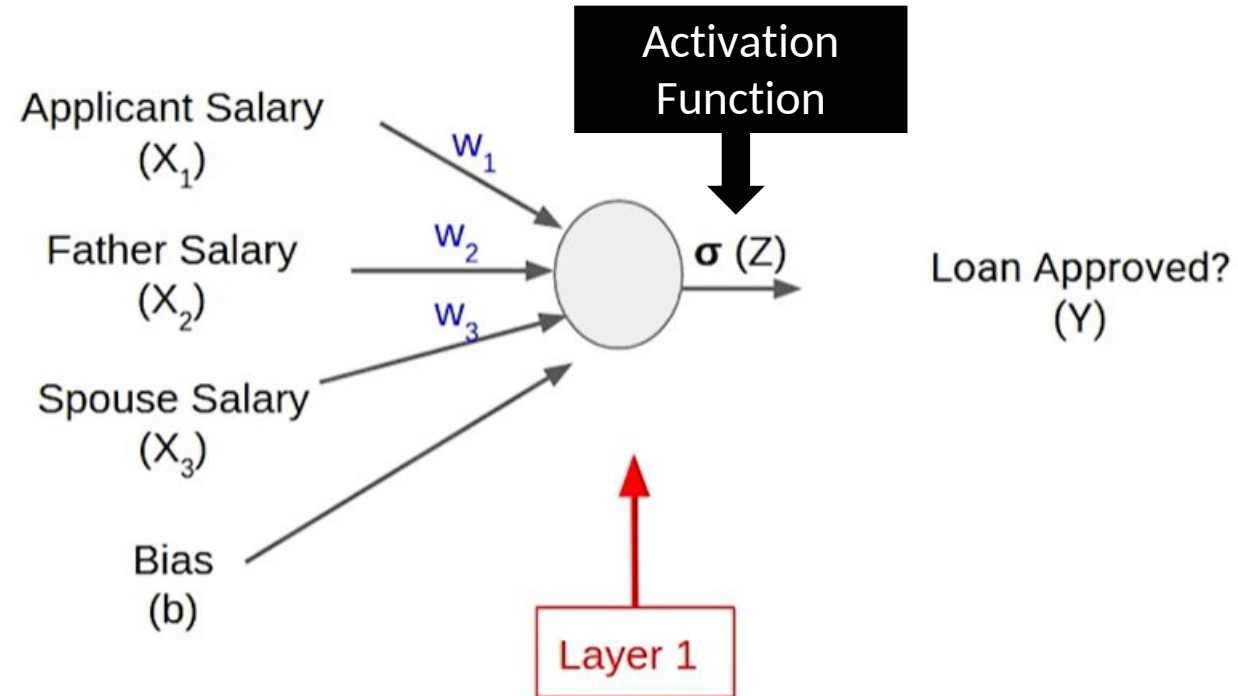
$$\text{Sum of inputs} = X_1 * w_1 + X_2 * w_2 + X_3 * w_3$$

$$Z = X_1 * w_1 + X_2 * w_2 + X_3 * w_3 + b \text{ (bias)}$$

$$\hat{Y} \text{ (output)} = \text{step} (Z)$$

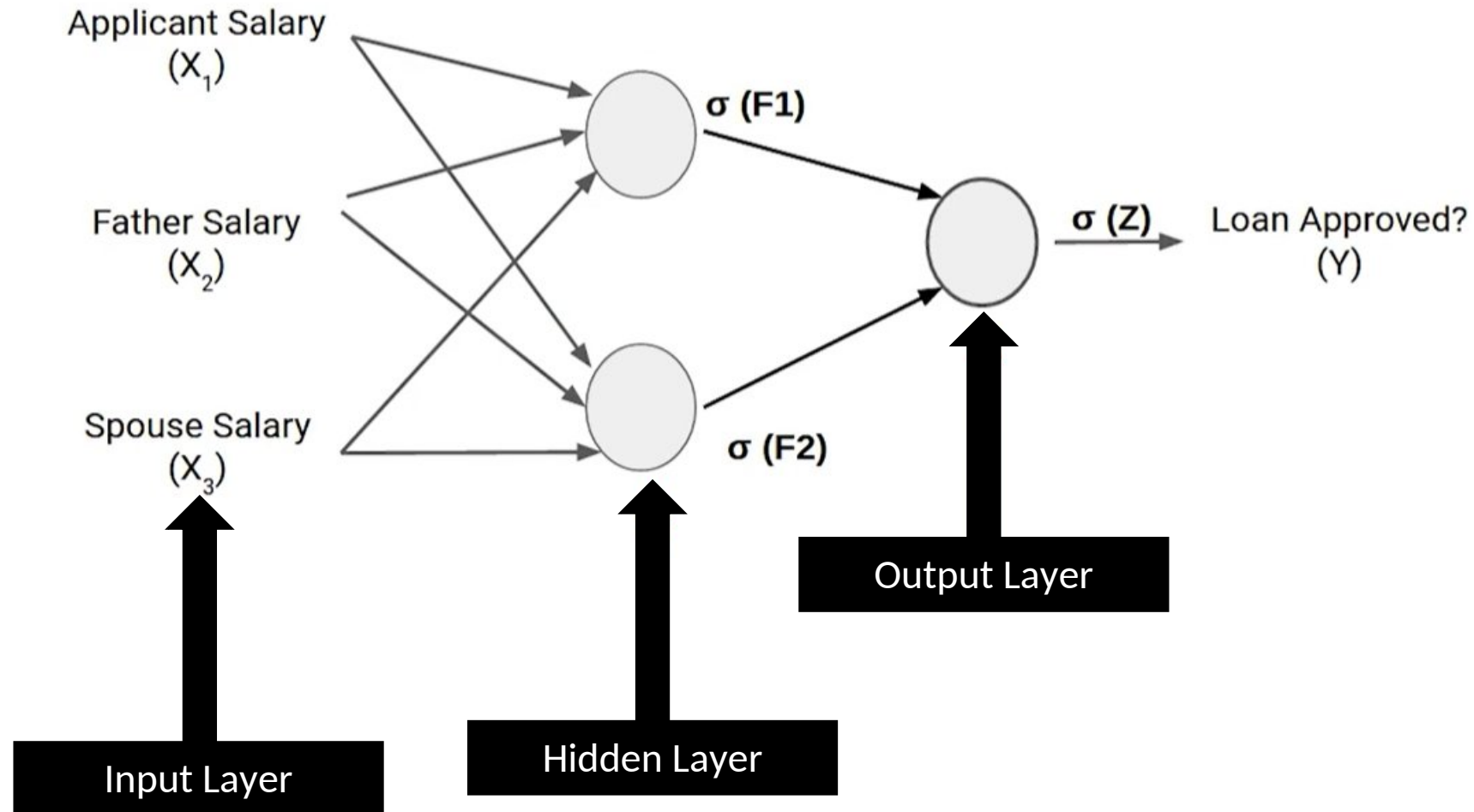
Single Layer Perceptron

Here we have only one neuron in the hidden layer.



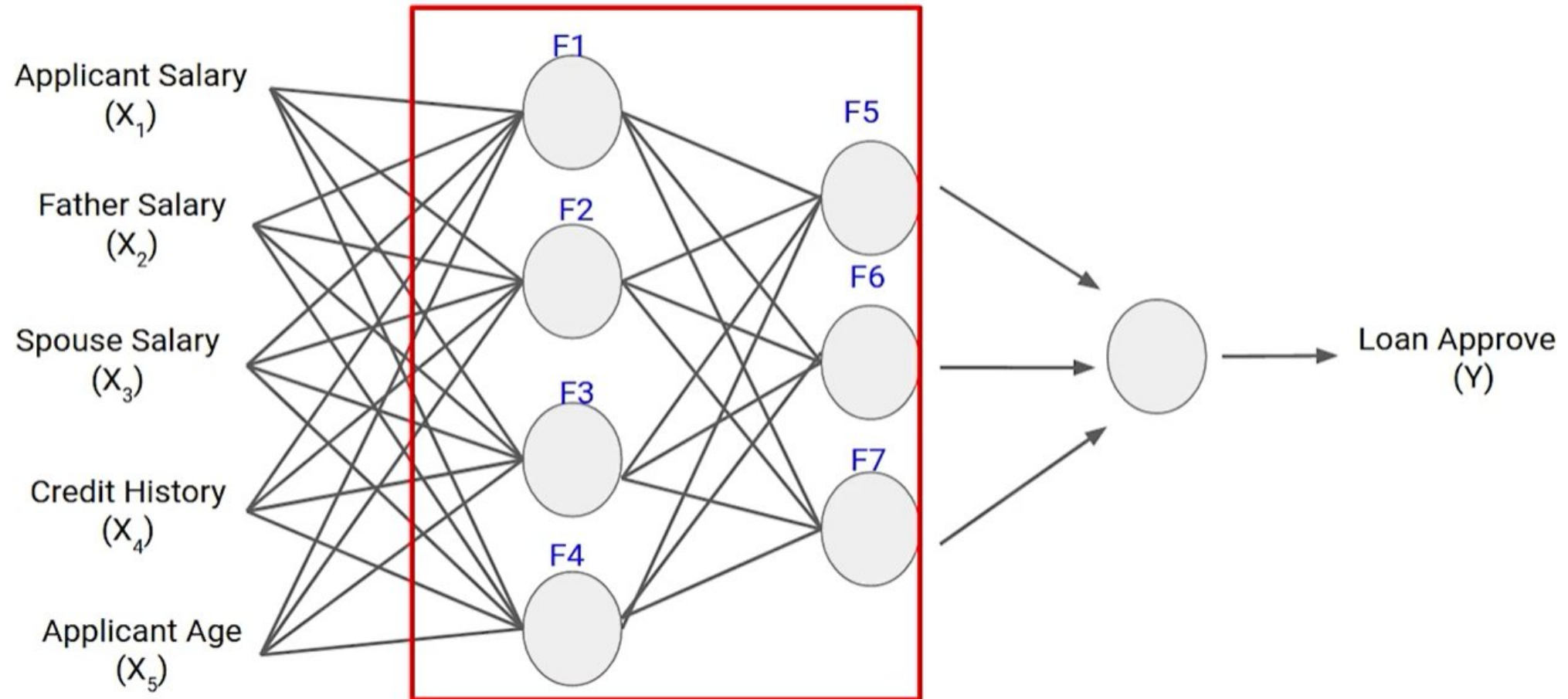
Multi Layer Perceptron

We can have more than one neuron in the hidden layer.

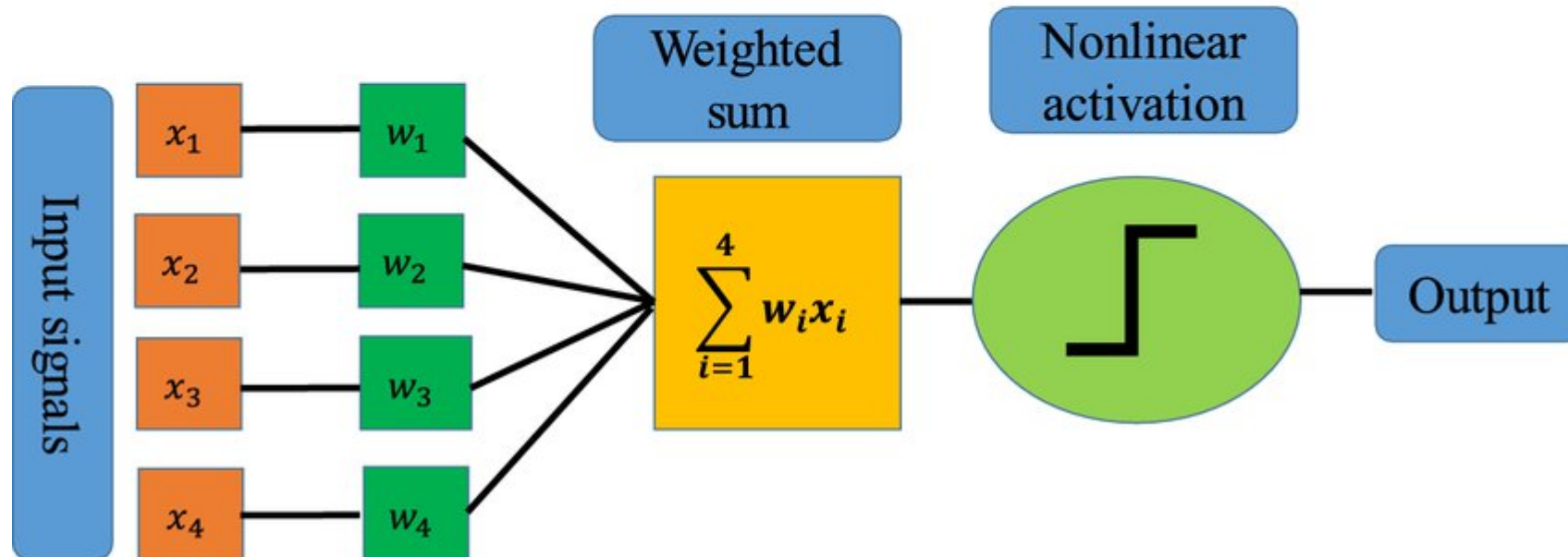
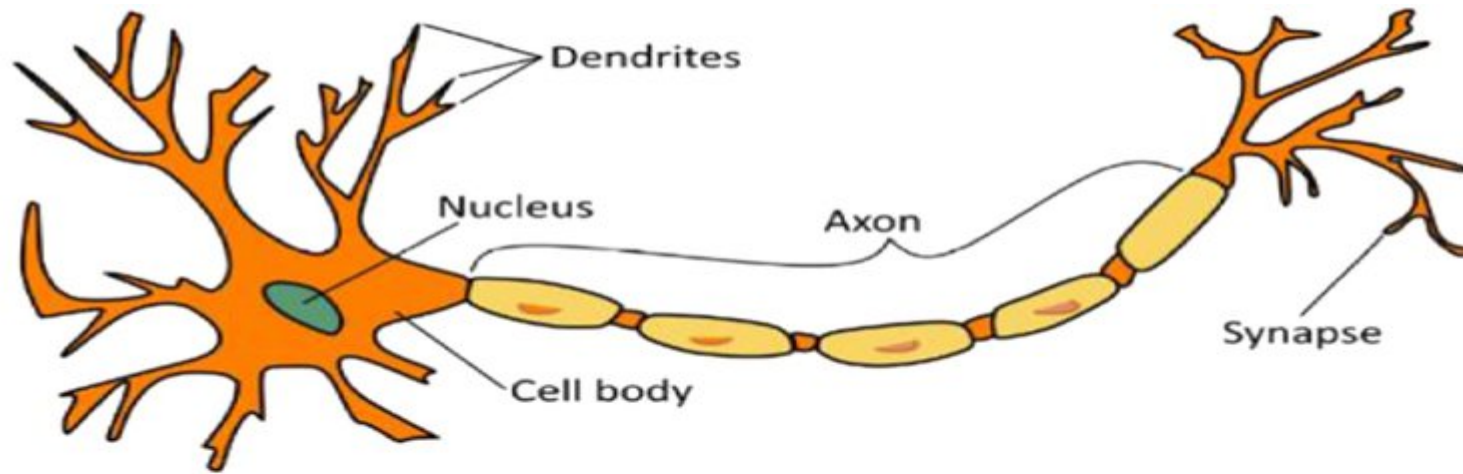


Multi Layer Perceptron

More than one layers can be added.



Biological Neuron vs Artificial Neuron



Biological Neuron vs Artificial Neuron

Biological Neuron	Artificial Neuron	Functionality
Dendrites	Input Features ($x_1, x_2...$)	Dendrites receive signals from other neurons, just like how an artificial neuron receives input values (features) from data.
Synaptic Weights	Weights ($w_1, w_2...$)	Synapses strengthen or weaken signals; similarly, each input in an artificial neuron is multiplied by a weight that adjusts the signal's importance.

Biological Neuron vs Artificial Neuron

Biological Neuron	Artificial Neuron	Functionality
Cell Body (Soma)	Summation Function	The soma integrates the signals, similar to how an artificial neuron computes the weighted sum: $z = \sum(w_i \cdot x_i) + b$.
Axon Hillock	Activation Function	The axon hillock decides whether to fire or not. In an artificial neuron, this is handled by an activation function (e.g., ReLU, Sigmoid, Step).

Biological Neuron vs Artificial Neuron

Biological Neuron	Artificial Neuron	Functionality
Axon	Output Value	The axon transmits the output signal. In the artificial neuron, the result of the activation function becomes the output to the next neuron/layer.
Synapse (at Axon End)	Connection to next layer	Just like a synapse sends signals to other neurons, the output of one artificial neuron becomes input to neurons in the next layer.